

Research Plan: High-resolution Coral Geodesy in the Solomon Islands

Overview and Preliminary Findings

The primary objective of this research is to introduce a new tectonic proxy derived from stable isotope records in coral skeletons. Preliminary results indicate that the stable carbon isotope ratio of C^{13}/C^{12} ($\delta^{13}C$) is highly sensitive to the relative sea level (RSL) at the precise location of the coral's growth. To test this, we analyzed 575 samples to track the temporal variation of $\delta^{13}C$ in a specific coral colony from the New Georgia group in the Solomon Islands. Our findings reveal significant tectonic trends across decadal timescales, alongside climate-driven fluctuations at various frequencies.

Phase I: High-Resolution Validation and Age Modeling

Our immediate next step involves the high-resolution sampling and analysis of a "twin coral" a colony situated immediately adjacent to the original study site. We plan to increase the sampling rate to capture finer temporal details and also, we plan to analyze additional isotopes alongside $\delta^{13}C$ and $\delta^{18}O$ to isolate climatic variables from tectonic signals. This is essential for establishing a precise Bayesian age model and decoupling environmental noise from true vertical deformation.

Phase II: Regional Scaling and Slip Modeling

Following validation, the research will expand to include multiple corals collected within a 50 km radius. By synthesizing these localized records, we aim to construct a robust local slip model to reconstruct the historical behavior of the underlying subduction interface.

Phase III: Geodetic Validation with NISAR

To bridge the gap between historical proxy data and modern neotectonics, we will integrate recently acquired NISAR (NASA-ISRO Synthetic Aperture Radar) data. This satellite imagery will allow us to validate the amplitudes and rates of deformation derived from the coral records against current surface deformation observed via Interferometric SAR (InSAR).